

Tick-Primed Hypersensitivity in Cosmetic and Aesthetic Procedures



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ALPHA-GAL SYNDROME (AGS)

SENSITIZED ≠ REACTIVE

Alpha-gal syndrome (AGS) necessitates a compatible history + α-Gal-specific IgE, IgE alone does not.

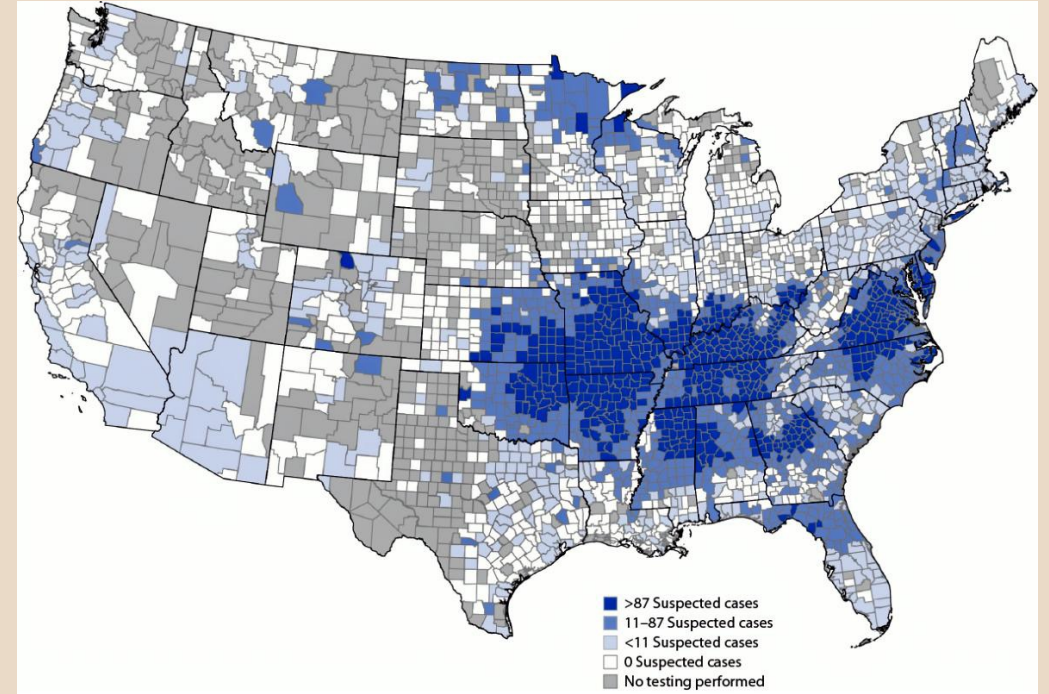
WHAT IT IS

IgE-mediated allergy to galactose-α-1,3-galactose (α-Gal), found in non-primate mammalian tissues and products^{3,9}

WHY IT MATTERS

Potential anaphylaxis risk: gelatin hemostats, collagen-containing fillers, mammalian grafts, catgut, and excipients^{3,5,9}

EPIDEMIOLOGY



Suspected AGS cases/million/year CDC/MMWR, 2017–2022¹²

90,018 suspected cases, 2017–2022.¹²
96,000–450,000 estimated affected since 2010.¹²
Seropositive ≠ disease: Arkansas 31.2%; Missouri 26.0%.¹⁰

REVIEW DESIGN

Literature published January 2010 through July 2026; final search 2 August 2026. Prioritized surgical/peri-op AGS.
Full methods and disclosures: QR.

LIMITS

No prospective cosmetic/aesthetic cohort or reaction registry. No validated product-specific IgE threshold or brand α-Gal assay.

ABBREVIATIONS

ADM acellular dermal matrix
AGS alpha-gal syndrome
AI artificial intelligence
CaHA calcium hydroxylapatite
CDC Centers for Disease Control and Prevention
HA hyaluronic acid
IgE immunoglobulin E
IV intravenous
kU/L kilounits per liter
MMWR Morbidity and Mortality Weekly Report
PLLA poly-L-lactic acid
PMMA polymethyl methacrylate
PPV positive predictive value
QR quick response code
UFH unfractionated heparin

02 • CLINICAL DECISION

SCREEN → STRATIFY → ACT

SCREEN	01	HISTORY • reactions + tick exposure ⁹	FOOD DATA ≠ PRODUCT CUTOFFS 2.0 kU/L / 0.75% total IgE: >95% PPV; >5.5 kU/L / 2.12%: ~95% modeled probability. <i>Not validated for cosmetic products.¹¹</i>
	02	ESTABLISH • diagnosis once: compatible history + α-gal IgE ^{3,9}	
STRATIFY	03	AUDIT • hemostats, fillers, grafts, sutures, Excipients ^{1,3,5,13}	ROUTE CHANGES TIMING Oral: delayed 3–6 h. Parenteral: may be immediate. Implant: prolonged exposure. Parenteral delivery bypasses gut Processing ^{3,5,9}
	04	VERIFY • species, source, excipients, lot ^{3,5,13}	
ACT	05	SUBSTITUTE • source-verified HA + non-gelatin options ^{3,9,13}	SOURCE + LOT MATTER Verify species, source, excipients, and lot. Formulations change; labels may be incomplete ^{3,5,13}
	06	PREPARE • epinephrine + reaction plan; Allergy if uncertain ^{5,9}	

03 • MATERIAL RISK

SENSITIZED ≠ REACTIVE • HISTORY + α-GAL IgE DEFINE AGS RISK

Evidence grade: documented reaction → clinical signal → mechanism → theory → no AGS-specific literature

EXPOSURE	EVIDENCE	SIGNIFICANCE
Gelatin hemostats GELFOAM • FLOSEAL ^{1,5,13}	DEMONSTRATED	Class-level perioperative anaphylaxis; not every brand.
Porcine UFH Cardiac-dose IV ^{4,13}	CLINICAL SIGNAL	Strongest with high-dose IV; route/dose matter.
Mammalian matrices STRATTICE • SurgiMend ^{3,13}	MECHANISTIC	Surgical precedent; no aesthetic cohort.
Collagen PMMA • catgut Bellafill • surgical gut ^{3,5,13}	THEORETICAL	Composition-based; no AGS reaction series.
Animal hyaluronidase[†] Amphadase • Vitrase ^{13,14}	THEORETICAL	Testicular extract ¹⁴ ; recombinant human HYLENEX = non–animal-extract alternative ¹⁶
Source-verified substitutes HA • CaHA • PLLA • lidocaine ^{2,3,9}	NO AGS-SPECIFIC LITERATURE	Other hypersensitivity mechanisms remain possible.

[†] **Hyaluronidase:** animal-derived ovine/bovine testicular products^{13,14} carry independent non-AGS allergy/anaphylaxis risk.¹⁵ Recombinant human HYLENEX is the non–animal-extract alternative.¹⁶

Neutrality: brands illustrate classes—not endorsement or brand-level α-gal assays.

Currency: examples current Aug 2026; verify current FDA label/IFU, species/source, excipients, and lot.¹³

SELECTED REFERENCES • FULL CITATIONS, ACCESSIBLE TEXT + AUDIO VIA QR

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DIGITAL COMPANION

tinyurl.com/AGS-FPRS

AUDIO + TRANSCRIPTS

